Entity-Relationship Model

INT191 Relational Database Concepts and Design

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What you will learn

- Database Design with Entity-Relationship Diagram (ERD)
- Basic Concepts of ER Model

Review

Database Design

Requirement Collection and Analysis

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1. Conceptual database design (Conceptual data model)

Entity Relationship Diagram

Independent of implementation details (applications, programming languages, target DBMS)

↓

2. Logical database design (Logical data model)

Tables and Constraints

Dependent of implementation details (applications, programming languages, target DBMS)

↓

3. Physical database design (Physical data model)

Describe how data is stored in order to access data efficiently

↓

Data files, Indexes, Partitions, Data block sizes

System Definition

- This stage involves **the fact-finding techniques:**
  - examining documentation
  - interviewing
  - observing the enterprise in operation
  - research
  - questionnaires
- Output is **requirement specification**

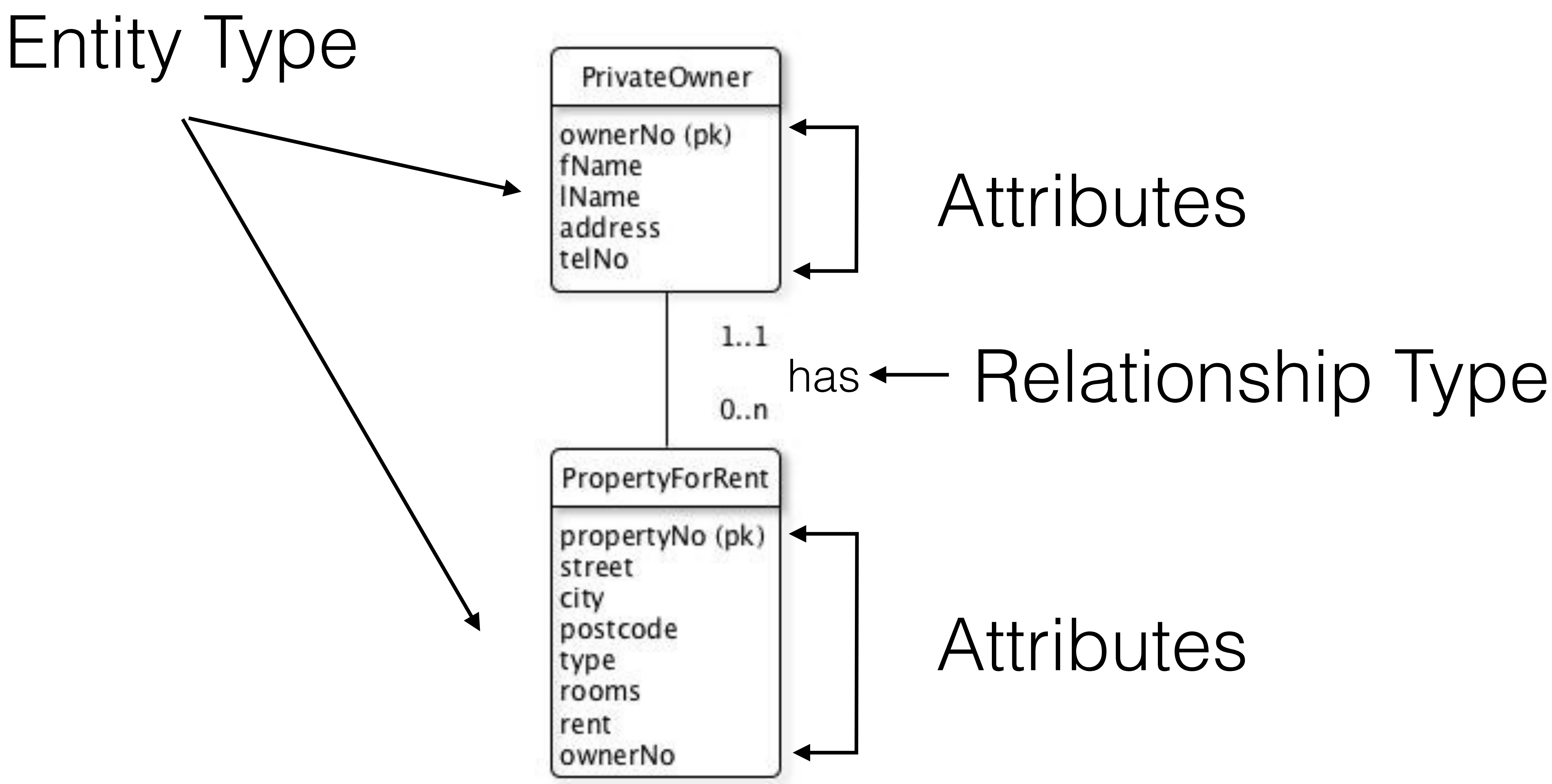
Approaches to Database Design

- **The bottom-up approach**
    - begins at the fundamental level of **attributes.**
    - the associations between **attributes are grouped into tables.**
    - suitable for **simple** databases with **small** number of attributes.
- **The top-down approach**
    - starts with a few **high-level entities.**
    - refinements to **identify** the associated **attributes.**
    - using the concepts of the **Entity-Relationship (ER) model** to identify entities and relationships.

Entity-Relationship (ER) Model

- ER Model
  - Is a top-down approach to database design.
  - Begins by identifying the important data (entities) and relationships between the data.
  - Then add more details (attributes and constraints).

ER Model


Entity Type
PrivateOwner
ownerNo (pk)
fName
lName
address
telNo
Attributes
1..1
has
0..n
Relationship Type
PropertyForRent
propertyNo (pk)
street
city
postcode
type
rooms
rent
ownerNo
Attributes

Concepts of the ER model

- **Entity Type**

  "A collection of entities (persons, places, events or things) of interest with the same properties.

  Entity type is represented by a rectangle"

- **Attributes**

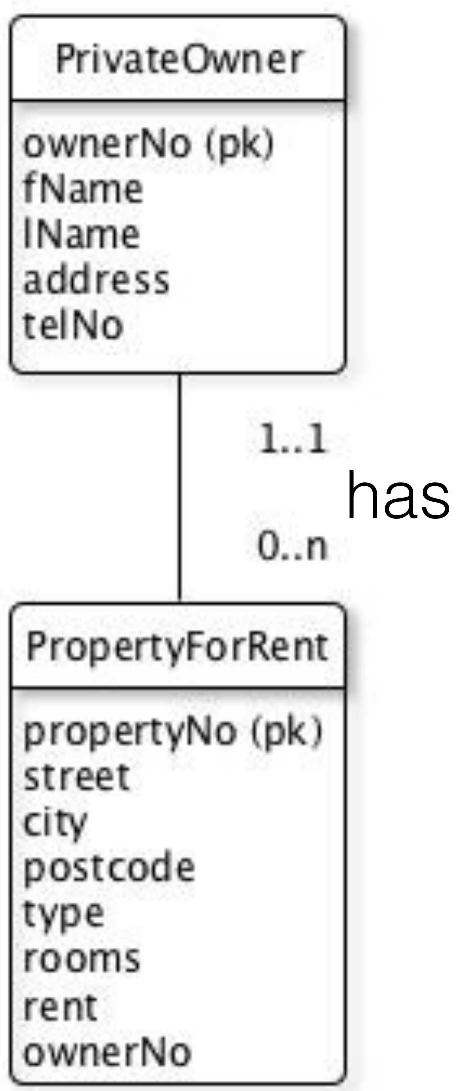
  "A property of an entity type or relationship.

  Each attribute has a data type that defines the kind of values and operations on the attribute"

- **Relationship Type**

  "A name association among entity types. The relationship represent a two-way association among entities"

ER Model vs. Instance Diagram



ER Diagram

Instance Diagram

Relationship Types

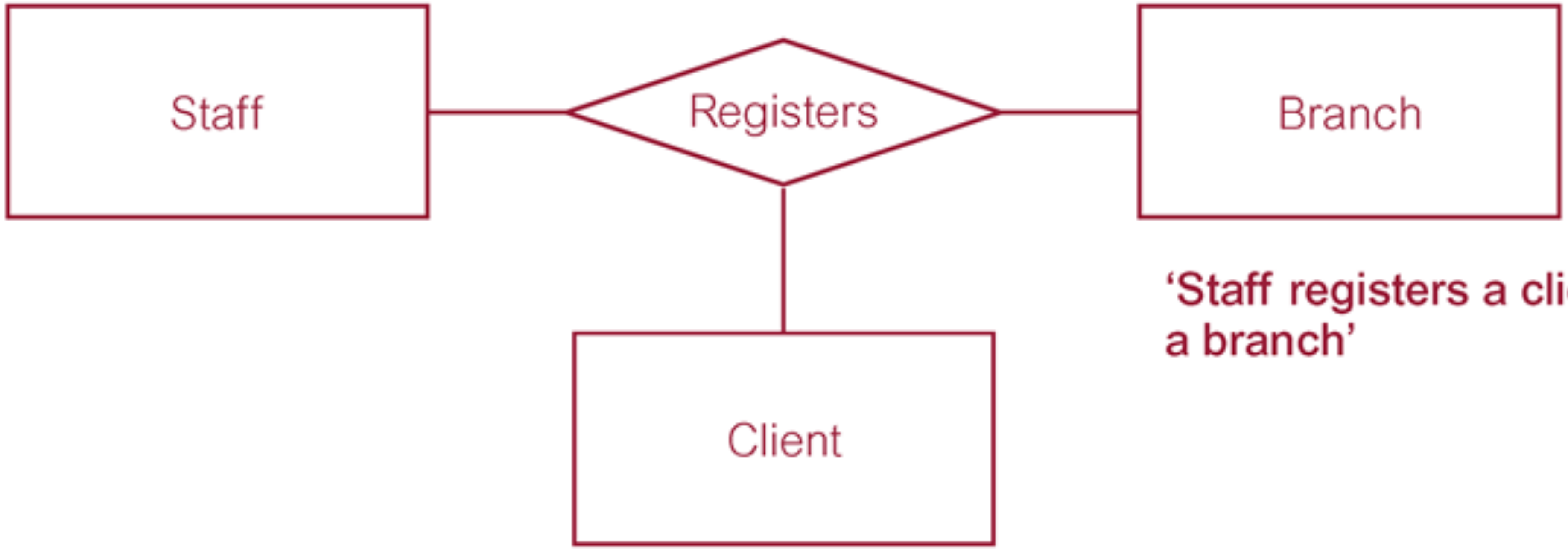
- **Degree** of a Relationship
  - Number of participating entities

- **Relationship Types**
  - Binary relationship (two) ***The most type***
  - Ternary relationship (three)
  - Quaternary relationship (four)
  - Recursive relationship (same entity)

Binary Relationship called POwns

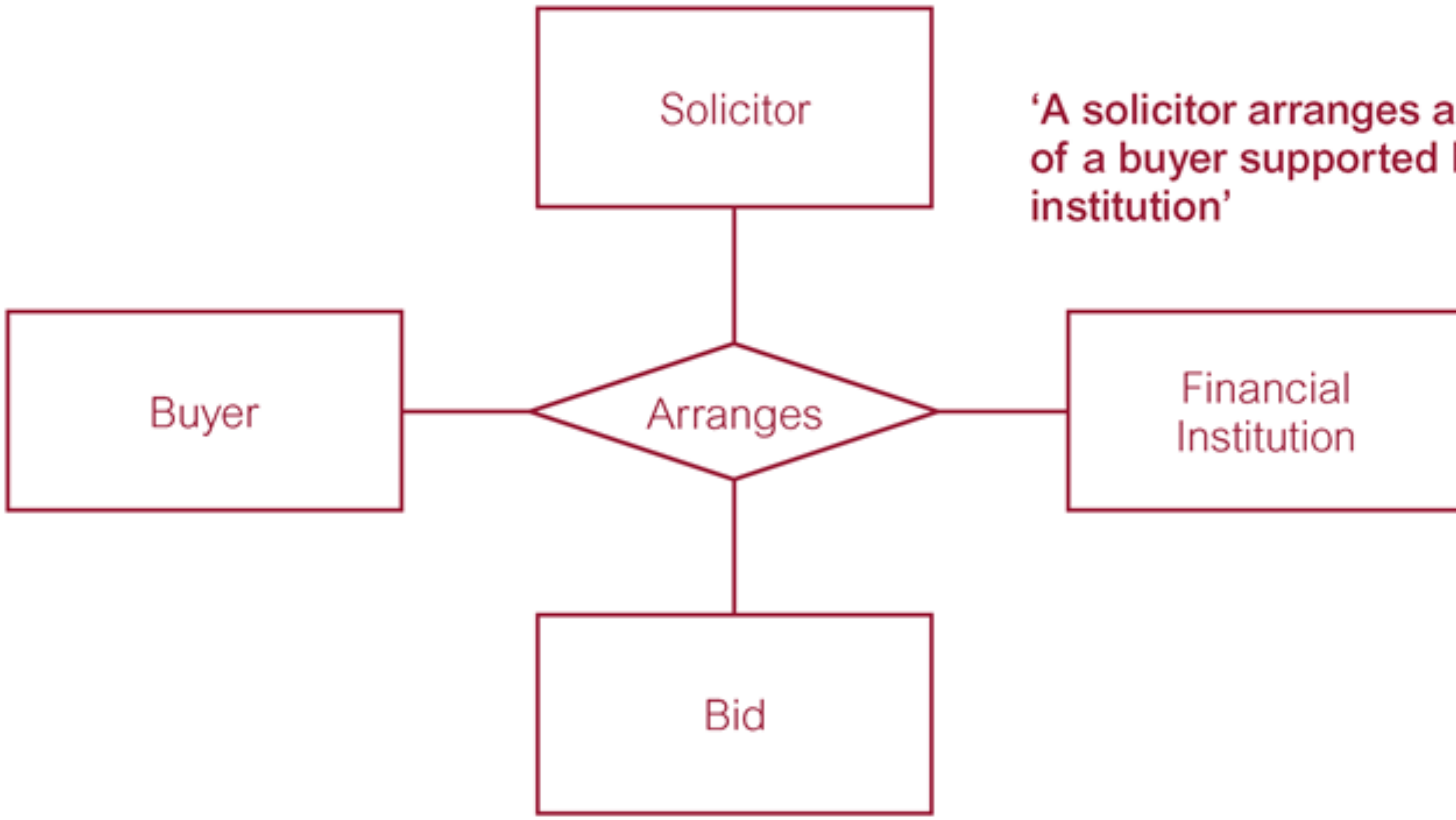
'Private owner owns property for rent'

Ternary Relationship called Registers



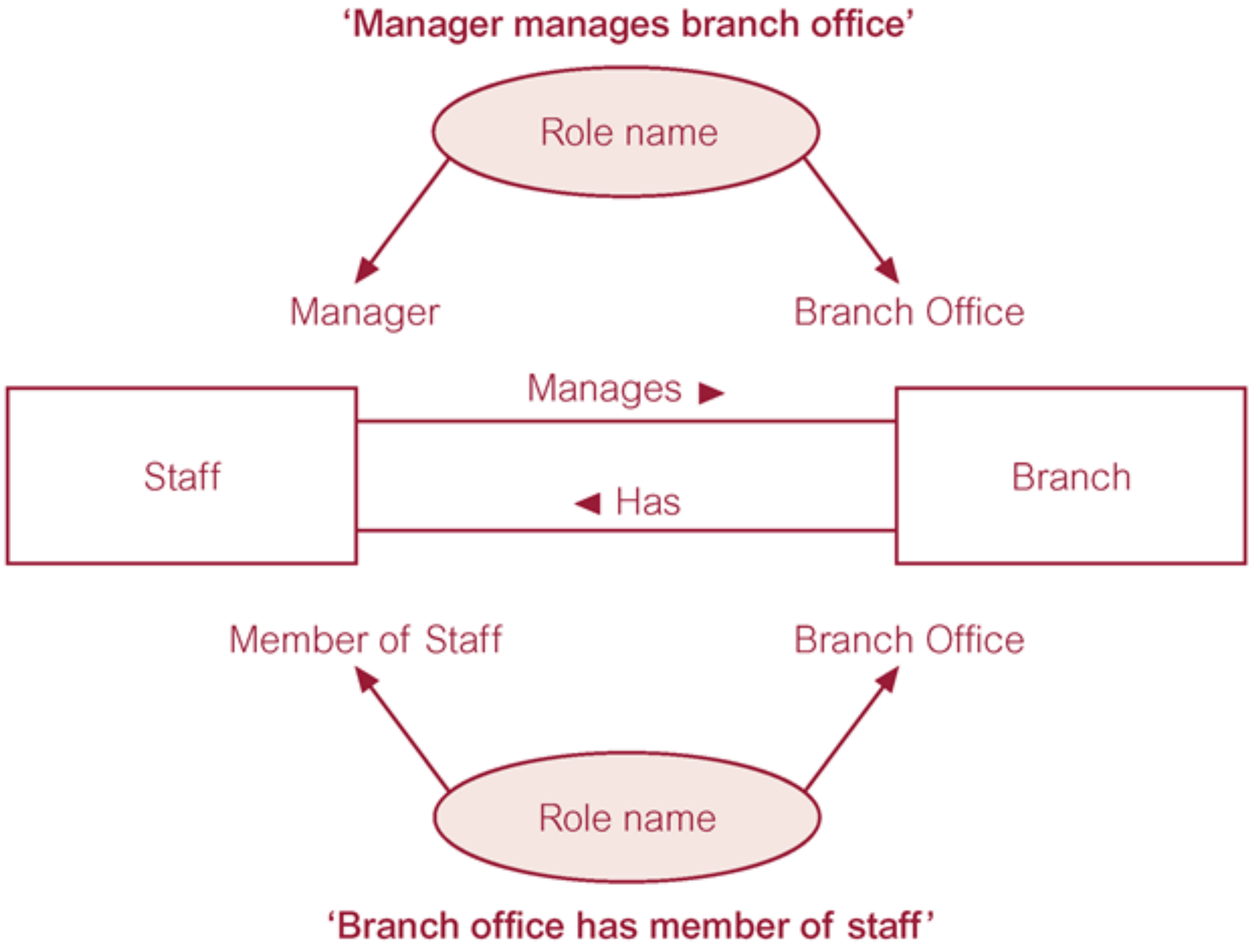
'Staff registers a client at a branch'

Quaternary Relationship called Arranges



'A solicitor arranges a bid on behalf of a buyer supported by a financial institution'

Entities associated through two distinct relationships with role names



Attributes

- **Attribute**

  "Property of an entity or a relationship type"

- **Attribute domain**

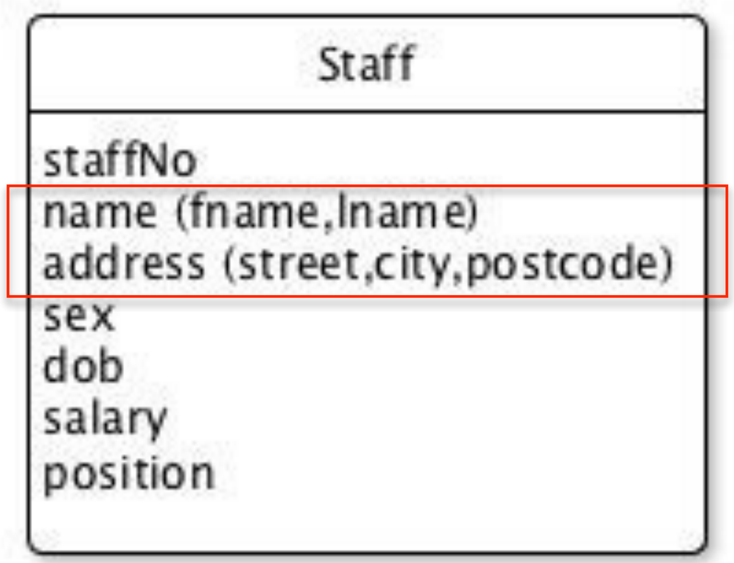
  "Set of allowable values for one or more attributes"

- **Single** vs. **Composite** attributes
- **Single-valued** vs. **Multiple-valued** attributes
- **Derived** attributes

Single vs. Composite Attributes

Single Attribute

"An attribute composed of a single component"



Composite Attribute

"An attribute composed of multiple components"

Single-valued vs. Multiple-valued Attributes

Single-valued Attribute

"An attribute that holds a single value"

Course
courseNo courseName credit requisiteCourseNos

Course Table

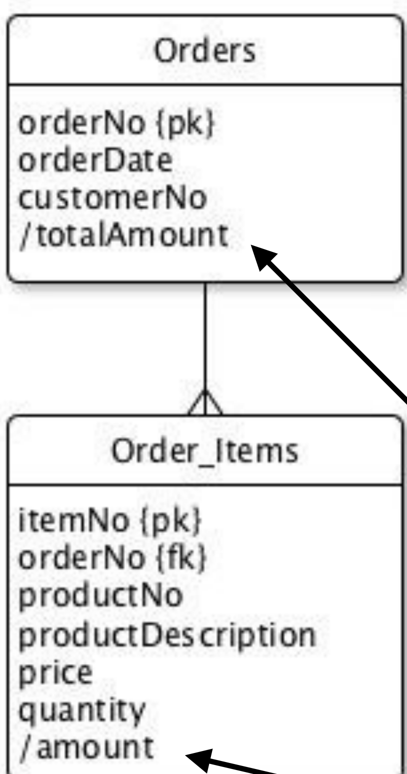
courseNo	courseName	credit	requisiteCourseNos
INT102	Computer Programming	3	
INT105	Computer Programming II	3	INT102
INT203	Information Management I	3	
INT207	Information Management II	3	INT102, INT203

Multiple-valued Attribute

"An attribute that holds multiple values"

Derived Attributes

"An attribute that represents a value that is derivable from the value of a related attribute or set of attributes, not necessarily in the same entity type"



orderNo	orderDate	customerNo	totalAmount
1	1-Jan-15	A100	20000

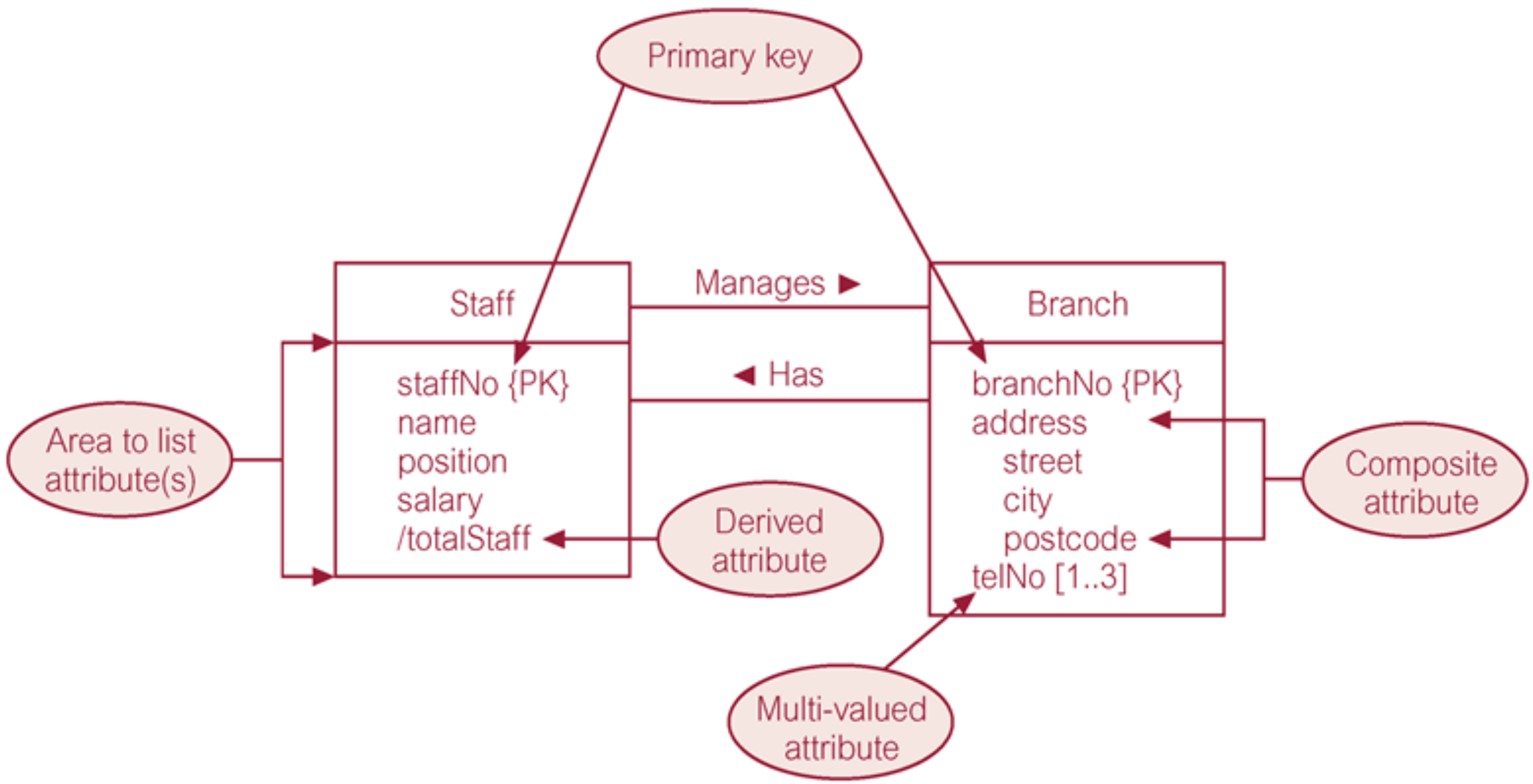
totalAmount=9000+11000

ItemNo	orderNo	productNo	productDescription	price	quantity	amount
1	1	1001	Monitor 17"	4500	2	9000
2	1	3005	Mainboard	5500	2	11000

Derived attributes

amount =price * quantity

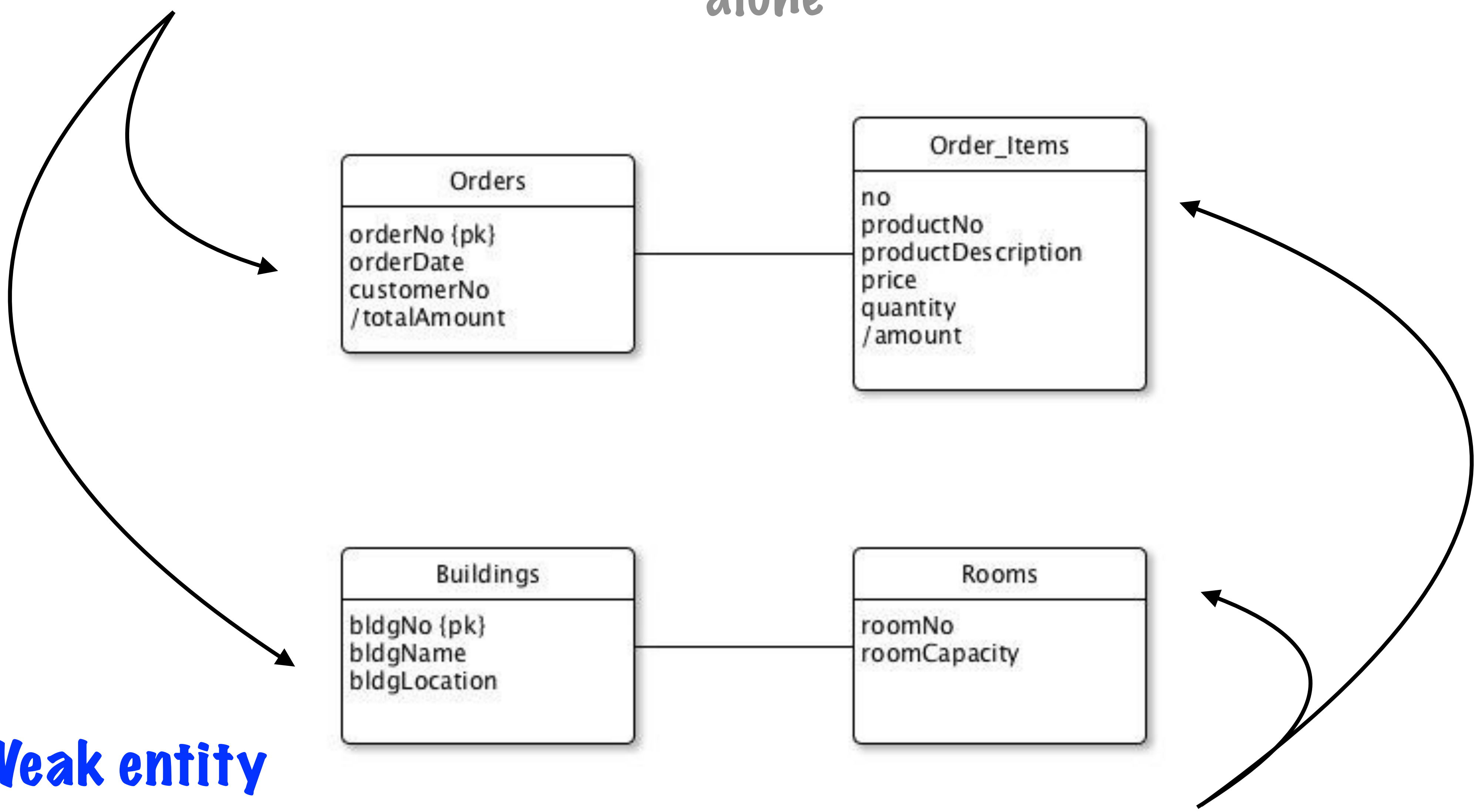
Diagrammatic Representation of Attributes



Strong vs. Weak Entity Types

- **Strong entity**

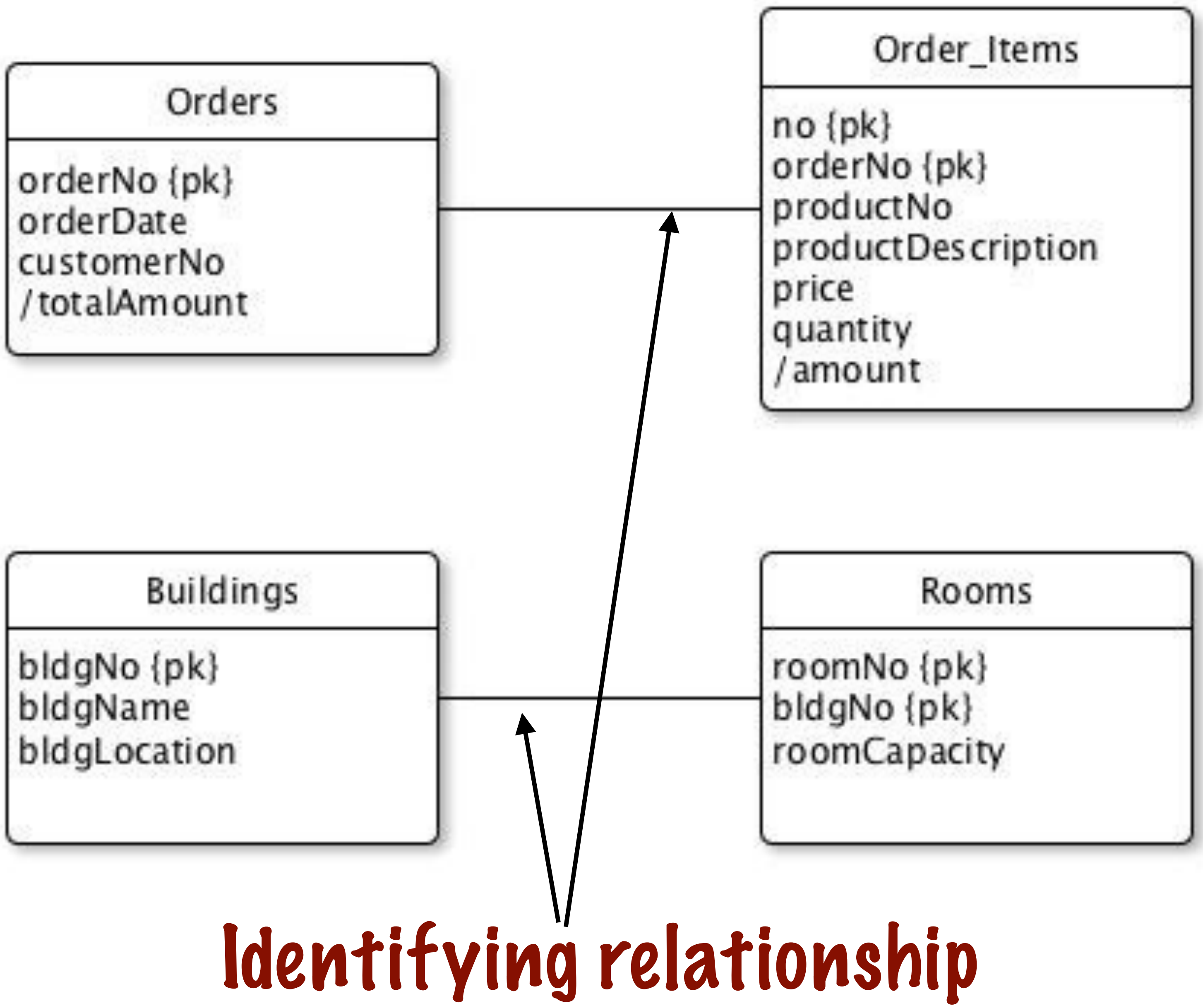
  "An entity that can be uniquely identified by its primary key attribute alone"



- **Weak entity**

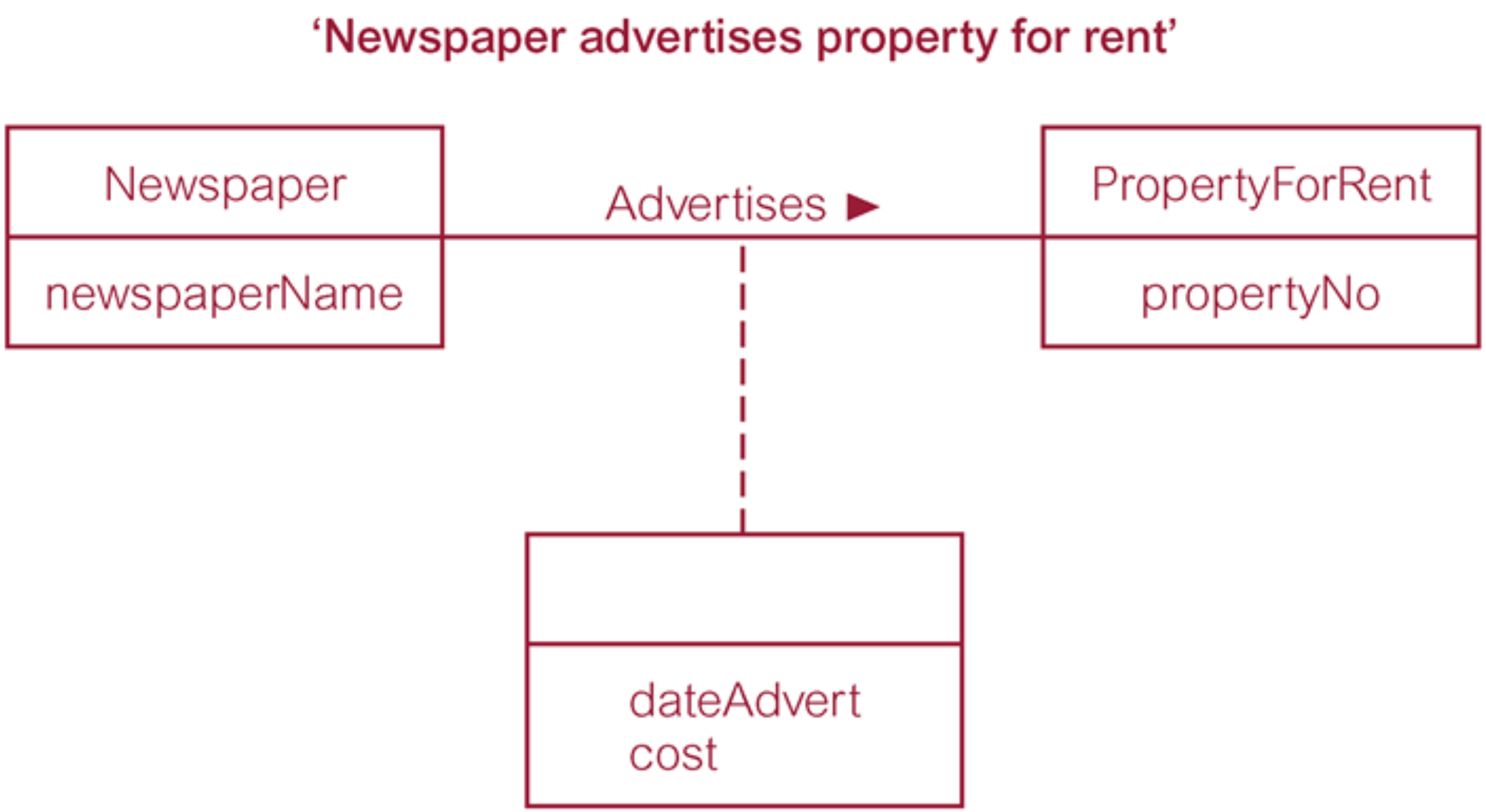
  "An entity that cannot be uniquely identified by its attributes alone"

Convert Weak Entity to Strong Entity



"The key of strong entity type is required to uniquely identify instances of the weak entity"

Relationship with Attributes



Structural Constraints

- **Constraints**
  - Should reflect the restrictions on the relationships as perceived in the "real world"
- **Example**
  - A property for rent must have an owner.
  - Each branch must have staff.

Multiplicity

- Main type of constraint on relationships is called **multiplicity**
- **Multiplicity**

  "The number (or range) of possible occurrences of an entity type that may relate to a single occurrence of associated entity type through a particular relationship"

- It is used to **identify** how the entities are related
- It represents policies (call **business rules**) defined by the user or company

Summary of Multiplicity Constraints

Alternative ways to represent multiplicity constraints	Meaning
0..1	Zero or one entity occurrence
1..1 (or just 1)	Exactly one entity occurrence
0..* (or just *)	Zero or many entity occurrences
1..*	One or many entity occurrences
5..10	Minimum of 5 up to a maximum of 10 entity occurrences
0, 3, 6–8	Zero or three or six, seven, or eight entity occurrences

Components in Multiplicity

- **Multiplicity** consists of:
  - **Cardinality**

    “The maximum number of possible relationship occurrences for an entity participating in a given relationship type”

  - **Participation**

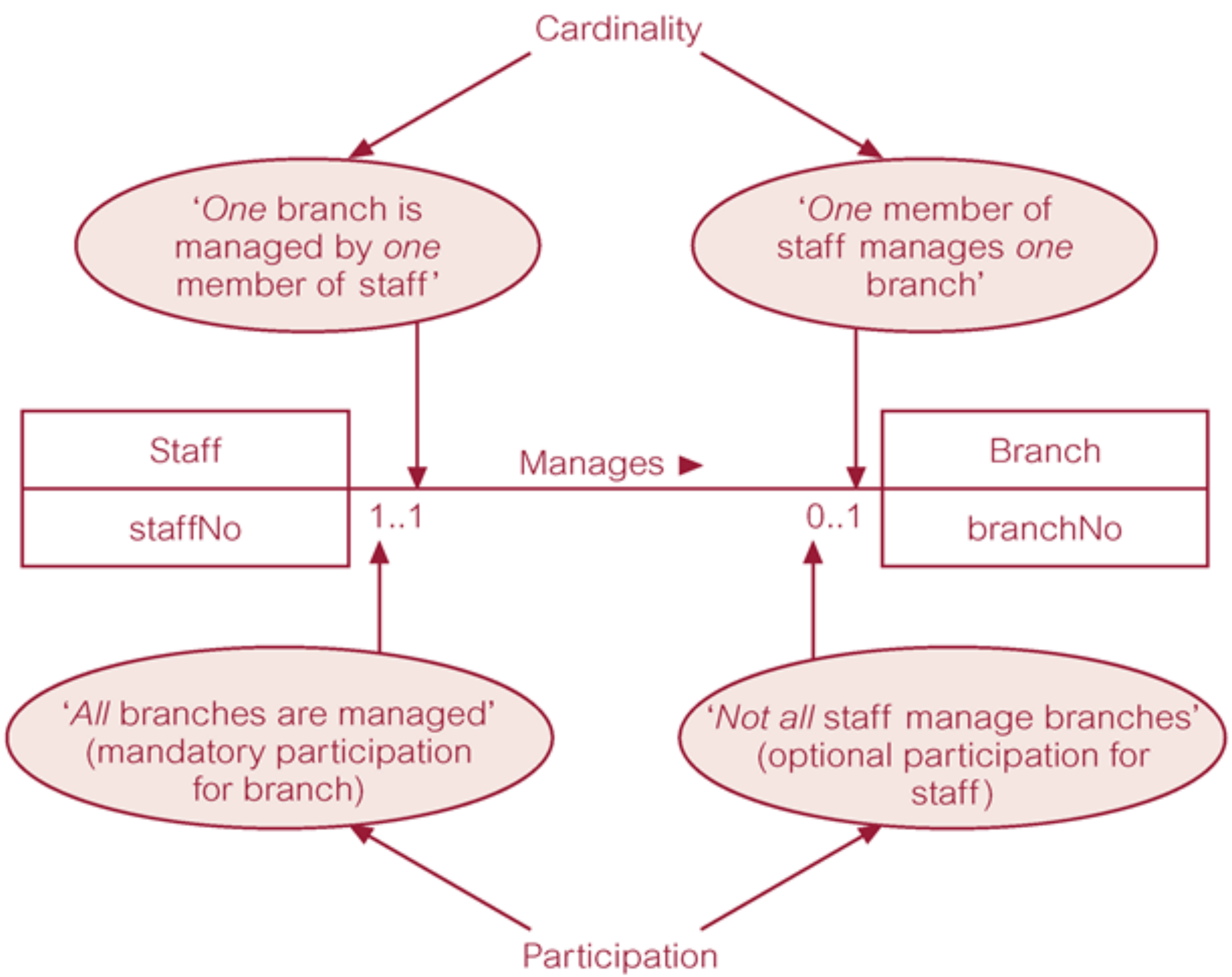
    “Determine whether all or only some entity occurrences participate in a relationship”

min .. max

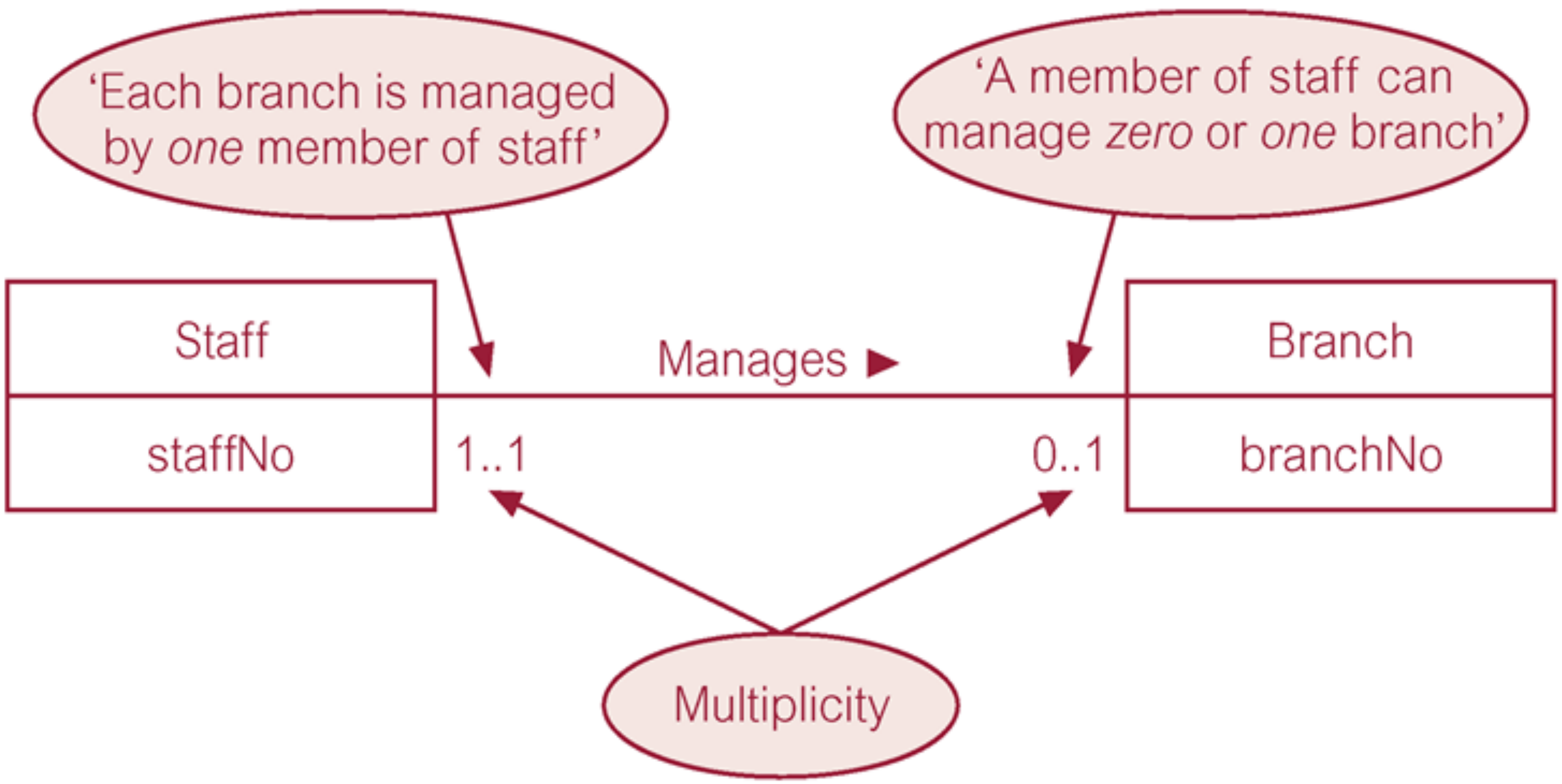
participation (0 or 1)

cardinality (1…n)

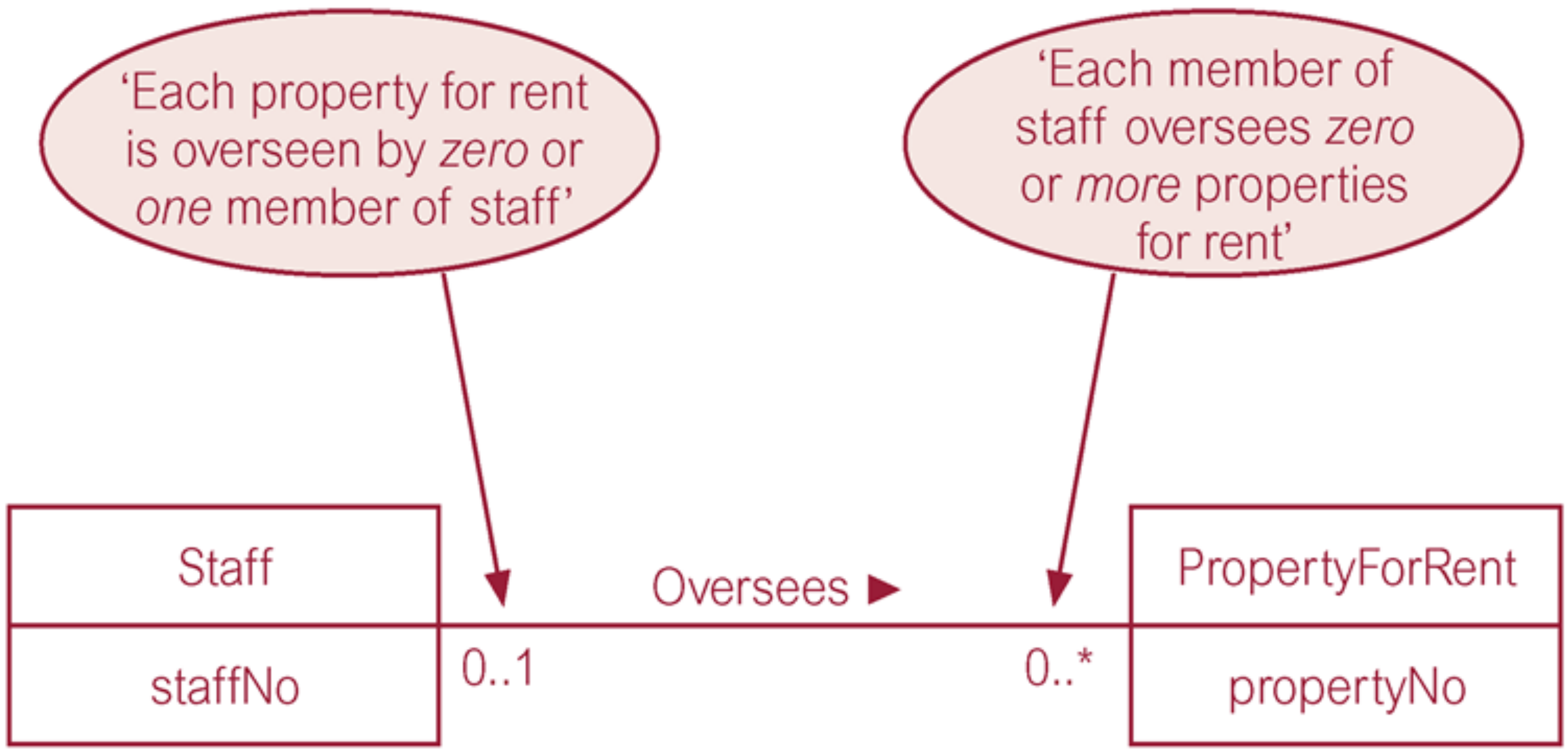
Multiplicity as Cardinality and Participation Constraints



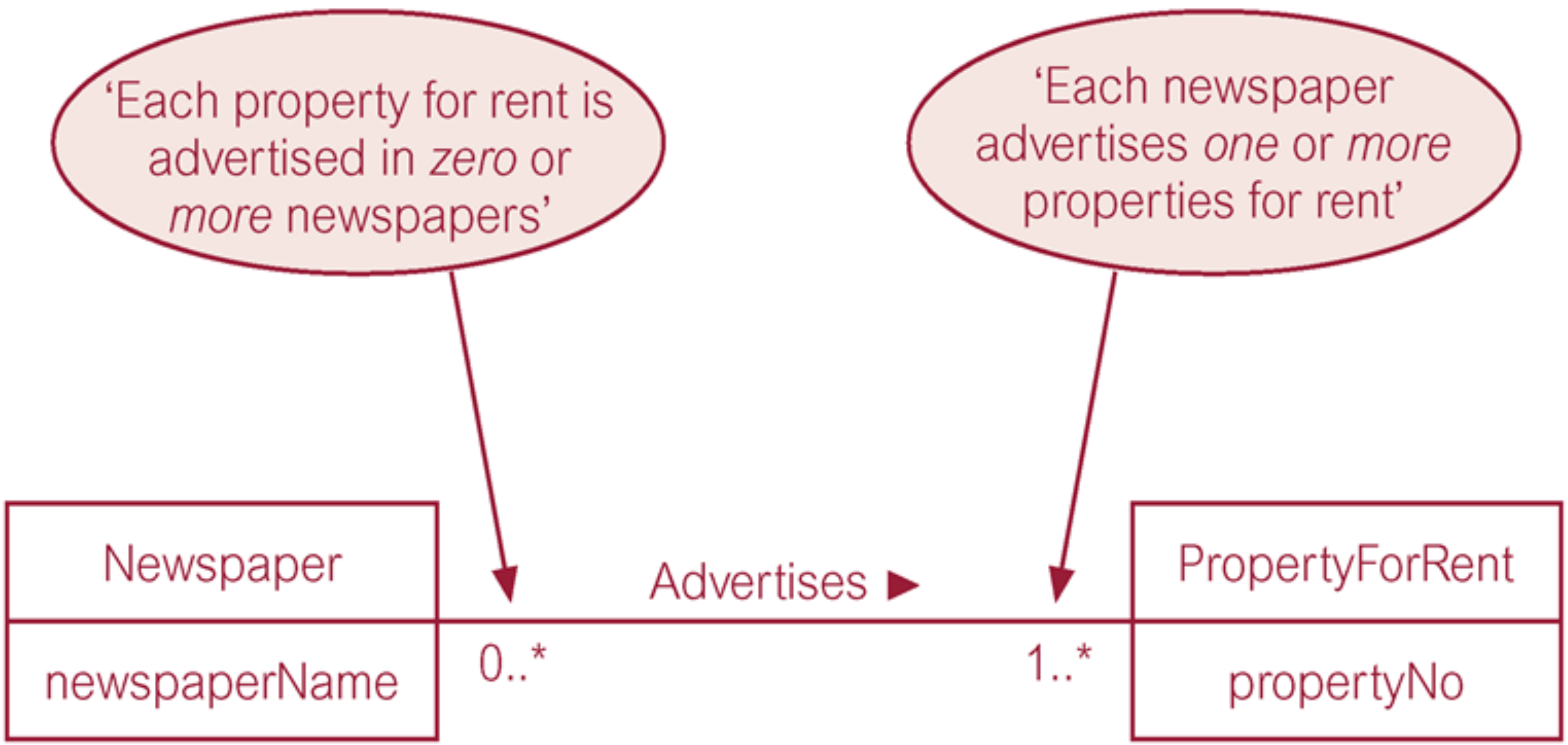
One-to-One Relationships (1:1)



One-to-Many Relationships (1:*)



Many-to-Many Relationships (*:*)



ER Diagram of Staff User Views of Dreamhome